

Automated Flower Classification



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DS4002 Prototyping
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Project Details

Research Question – How accurately can a CNN-based feature extraction classify flower images based on features such as petal shape and color?

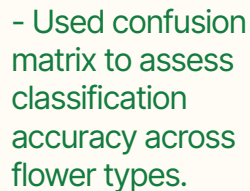
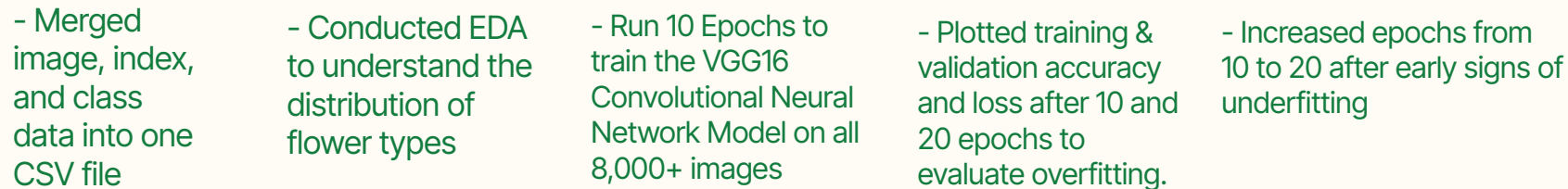
Motivation – Recognizing and protecting diverse flower species plays a crucial role in helping our ecosystem thrive.

Modeling Approach – Classify flower species by training a VGG16 Convolutional Neural Network (CNN) model on 8,000+ images of 102 different flower classification types.

Data Acquisition

- Downloaded 3 datasets from Oxford
 1. JPG folder of 8,000+ flower images
 2. Image ID and index
 3. Index and flower name
- The image dataset contains 102 flower classes, and each class consists of 40-258 images
- Kept the background for environmental context
- Merged the data into one final dataset → 'Final_Dataset.csv'

Filename	Name of JPG file
Path	File path
Image	Flower image, expressed as pixel values
Image_id	A unique numerical value corresponding to each individual image
Index	A numerical value corresponding to the flower classification
Name	The name/classification of the flower itself, which corresponds to the index



Tricky Analysis Decision

Choosing Number of Epochs:

- Too few → underfitting; too many → risk of overfitting and longer queue times on Rivanna.
- Settled on 20 epochs after evaluating training curve trends from 10.

Resource Constraints:

- GPU queue delays on Rivanna made testing multiple models/epochs difficult.
- Prioritized training depth over model experimentation.

Model Architecture Selection:

- Considered simpler CNNs vs. transfer learning models.
- Chose VGG16 for strong performance on image classification tasks with limited tuning.



Bias and Uncertainty

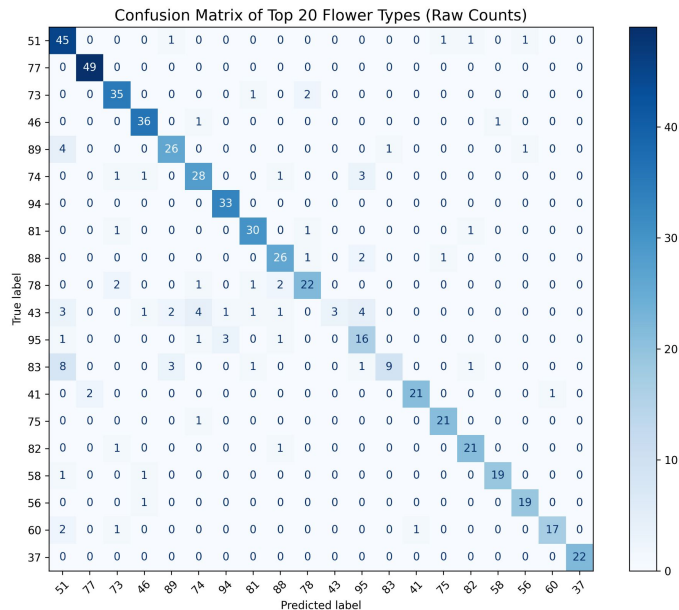
- Unequal distribution of flower types
 - There are significantly more passions flowers and petunias
 - The model might perform worse on underrepresented flower types
- Since we opted to include the background, potential obstructions and/or variation in image quality may impact the models' ability to identify key features of flower classes
- The CNN model could potentially focus on unimportant features such as background, lighting etc.

Classification Performance

- Trained for 20 epochs using VGG16 on over 8,000 flower images.
- Validation accuracy steadily improved from ~49% to 58%, with no clear signs of overfitting.
- Loss decreased consistently, indicating good convergence behavior.
- Small improvements after epoch 15 suggest the model was approaching its performance ceiling.

```
Epoch 15/20
205/205 [=====] - 373s 2s/step - loss: 2.7248 - accuracy: 0.2580 - val_loss: 1.7287 - val_accuracy: 0.5653
Epoch 16/20
205/205 [=====] - 380s 2s/step - loss: 2.6828 - accuracy: 0.2719 - val_loss: 1.7760 - val_accuracy: 0.5311
Epoch 17/20
205/205 [=====] - 379s 2s/step - loss: 2.6989 - accuracy: 0.2658 - val_loss: 1.6696 - val_accuracy: 0.5726
Epoch 18/20
205/205 [=====] - 384s 2s/step - loss: 2.6588 - accuracy: 0.2755 - val_loss: 1.7169 - val_accuracy: 0.5635
Epoch 19/20
205/205 [=====] - 390s 2s/step - loss: 2.6656 - accuracy: 0.2636 - val_loss: 1.7595 - val_accuracy: 0.5635
Epoch 20/20
205/205 [=====] - 405s 2s/step - loss: 2.6407 - accuracy: 0.2784 - val_loss: 1.5847 - val_accuracy: 0.5806
```

- Confusion matrix shows strong performance on the top 20 most frequent flower types.
- Some confusion remains between visually similar classes (e.g., classes 43, 95, and 83).



Next Steps

- Train model on more than 102 flower types/classes
 - Will not only improve the model's accuracy, but also increase its potential for real-world application
- Add more layers for higher accuracy → model will learn more complex patterns, which will help distinguish between similar flower classes
- Try a different model such as ResNet50

References

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- [7] ChatGPT (Version 4). *OpenAI* <https://chatgpt.com/>
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**THANK
YOU!**